Steering the Spectral Algebra of Attention Circuits in Transformer Pretraining

Li Hengyu[1]

[1]Institute for Solid State Physics, The University of Tokyo,
lihengyu@issp.u-tokyo.ac.jp

Abstract

Which algorithms a transformer acquires during training, and whether that choice can be steered, are central questions in mechanistic interpretability and for the prospect of models that are auditable by construction. A companion study of the QK operator's spectrum established that the positional encoding scheme sets a default spectral algebra for the attention heads implementing induction, and that this default is a preference rather than a constraint: banning a spectral channel is rerouted around at a measurable formation cost. We ask the constructive converse: whether *ban-free assistance*, either an initialization toward a chosen implementation or a weak population-level regularizer, can accelerate or select what forms. In pre-registered two-layer experiments the two levers separate cleanly: solution-specific initialization accelerates formation ($-28\%/-37\%$ across positional schemes, zero capability cost) but does not select the implementation; the weak regularizer selects (5/5 vs. 0/5 runs) but does not accelerate. The implementation flip that a ban forces at $2.9\times$ formation cost, assistance obtains at $1.3\times$. Both levers survive real-corpus pretraining at 160M parameters: a functional plant accelerates induction formation by 27–31% at capability parity, and weak assistance selects as deeply as a ban while remaining cheaper and faster within a dose window ($\lambda \leq 3$). A dose curve exposes the mechanism's limit, which we name *dose-defying concentration*: population pressure converts the population at the smallest dose, yet concentrates the default algebra into the load-bearing head. Steering learned circuits is thus cheaper than banning them, bounded by an active retention effect. Because the steered implementation evades weight-space fingerprints while remaining visible to propagator diagnostics, such interventions also remain auditable.

1 Introduction

Much of what makes transformer language models [19] capable arrives as *emergent circuits*: coordinated groups of attention heads, with the induction circuit behind in-context learning as the canonical example [5, 12, 20]. A substantial observational literature now reads such circuits out of trained weights and maps when they form during pretraining [2, 6, 13, 15, 17]. A younger line moves from observation to intervention: regularizing interpretable structure during pretraining [3], constraining the attention parametrization [4, 8], or shaping initialization [14]. The long-term prospect is models whose internal algorithms are, to a useful degree, chosen at training time rather than discovered afterwards: interpretability by construction rather than post hoc.

For attention, a concrete handle on "which algorithm" is the spectral structure of the QK operator $M = W_q^\top W_k$. Its symmetric/antisymmetric and complex-eigenvalue content separates head function [7, 11, 14], and the choice of positional encoding, whether rotary position embeddings (RoPE) [1, 16, 18] or learned absolute position embeddings (APE), determines which structure the previous-token heads feeding induction adopt. The first of two companion papers, hereafter the *QK-fingerprint* paper [9], showed across seven pretrained models that this choice is systematic (rotational under RoPE; content-like under APE and under the linear-bias scheme ALiBi), that the spectral signature consolidates after circuit formation, and that it is a default rather than a necessity: constrained training reroutes around any banned channel at a measurable cost, up to 2.9× slower formation. The second, hereafter the *propagator* paper [10], extends the account to the iterated attention propagator. Both are diagnostic: they price what training prefers and what prohibitions cost. This paper supplies the positive lever, assistance, and asks whether the same spectral economy that prices bans also enables steering.

Concretely: (Q1) does assistance toward a scheme's default algebra accelerate circuit formation? (Q2) does weak assistance toward the non-default algebra select which of two functionally-equivalent implementations forms, without bans? (Q3) how does the cost of steering compare to the cost of banning the same outcome? (Q4) do any of these survive real-corpus pretraining? Predictions were frozen before runs; we report falsified predictions with the same prominence as confirmations.

Contributions.

1. **Assistance is a lever, and its two ingredients separate** (§4): a solution-specific initialization accelerates circuit formation while a weak population-level regularizer selects the implementation. Initialization does not select; the regularizer does not accelerate. Algebra-class initialization and regularization fields do neither.
2. **Steering is cheaper than banning** (§4, §6): assistance reaches the same implementation a ban forces, at a fraction of the cost, both at toy scale and at 160M.
3. **Both levers transfer to real-corpus pretraining** (§6): at 160M/4B, a functional plant accelerates induction formation by 27–31% at capability parity, and weak assistance selects the non-default implementation as deeply as a ban while being cheaper and faster.
4. **A dose curve, and a mechanistic limit we name *dose-defying concentration*** (§6): population-mean pressure converts the population at the smallest dose, but the head that carries the function retains more default algebra as pressure grows. Per-head control is therefore a plant-side lever, not a penalty-side one. This turns the "suppress-generic / retain-functional" law of the QK-fingerprint and propagator papers from an observation into a dose-resolved, interventional result.

2 Related work

Circuit formation and its dynamics. Induction heads and their phase change are the canonical object of study [12]; formation has been dissected in toy models [2, 13, 15] and tracked across public checkpoints and scale [17]. This line is observational: it watches circuits form. We intervene on formation.

Constraining structure during training. The closest precedents constrain or regularize an interpretable property while training. Chen et al. [3] regularize a syntactic property during MLM pretraining; symmetric dot-product BERT trains competitively [4]; shared-QK attention is an architectural constraint [8]; and Saponati et al. [14] report that symmetric initialization can speed encoder training. These are constraints or architecture choices. Our levers are ban-free and, crucially, decomposed: we separate the initialization from the penalty and show they do different jobs.

The spectral algebra of QK. The symmetric/antisymmetric decomposition of the QK matrix [14] and per-head complex eigenvalues of the attention interaction [7] describe learned structure; the

QK-fingerprint paper [9] anchors it behaviorally and shows the positional scheme sets the default spectral algebra, reroutable at a measurable cost, while the propagator paper [10] analyzes the iterated propagator. This paper supplies the constructive converse those papers point to: not what training prefers, but what it can be persuaded to do, and at what price.

3 Method

Operators, metrics, and implementation classes. For a head with projections W_q, W_k, the pre-softmax score is the bilinear form $x_i^\top M x_j$ in the QK operator $M = W_q^\top W_k$ [5]. Under rotate-half RoPE [16], M decomposes into per-frequency complex sub-operators M_t, and the relative-position asymmetry of the score kernel is carried entirely by their imaginary parts: $\mathrm{score}(\Delta) - \mathrm{score}(-\Delta) = \sum_t 2\sin(\Delta\theta_t)\, n^\top \mathrm{Im}(M_t)\, n$ [9]. Two weight-only statistics summarize a head's implementation: `rope_imag_frac`, the share of QK mass in the rotary phase channel, $\sum_t \|\mathrm{Im}(M_t)\|_F^2 / \sum_t \big(\|\mathrm{Re}(M_t)\|_F^2 + \|\mathrm{Im}(M_t)\|_F^2\big)$, and `dir_frac` $= \|M_A\|_F / \|M\|_F$, the antisymmetric fraction of the static operator. A previous-token head is *phase-carried* when `rope_imag_frac` sits near the free-training default (≈ 0.5) and *cos-only* when `rope_imag_frac` ≤ 0.10; under APE the analogous classes are read from `dir_frac` (thresholds 0.35/0.10). All thresholds were pre-set from the QK-fingerprint paper's bimodal distributions before any run reported here. We reuse the QK-fingerprint paper's constrained-training harness: 2-layer attention-only models (d=128, 4 heads, d_k=32) trained from scratch on a per-sequence random-map induction task (successor prediction requires finding the previous occurrence of the current token and copying), n=5 seeds, formation step defined as the first eval crossing the induction-CE threshold (verbatim from that paper). Free (unassisted) arms reproduce its numbers exactly (600±0 APE, 940±55 RoPE). Assistance takes three forms, all of them ban-free: (i) *solution-init*, which initializes the seed heads' QK factors to carry the target implementation's positional offset (the Δ=−1 phase θ under RoPE; a symmetric nearby-position plant under APE); (ii) *algebra-init*, which installs only the target's algebra class (random phases, or a skew plant) without the specific offset; and (iii) *assist-reg*, a bounded complement penalty that pushes the population's mean spectral share toward the target, with detached denominators. Placebos apply the cross-scheme algebra (RoPE-style under APE and vice versa) and a random orthogonal plant. Implementation class is read out by the QK-fingerprint paper's weight batteries (`rope_imag_frac` for RoPE, `dir_frac` for APE), with the pre-set thresholds of §3. All outputs and per-run tables are released.

4 Results: toy scale

(1) Assistance accelerates, but only as solution-specific initialization. Figure 1 and Table 1: solution-init speeds formation in both schemes (RoPE 940→680, −28%; APE 600→380, −37%; one-sided exact MWU p=0.004, rank-biserial +1.0), at zero capability cost (final induction CE unchanged; the solution-init arms end with higher previous-token scores). Our primary pre-registered prediction held that initializing the algebra class would suffice; that prediction is **falsified as frozen**. Algebra-init is non-significant in both schemes (RoPE q_{BH}=0.16 with a suggestive rank-biserial of +0.64; APE null at the eval-grid resolution). Placebos are null, passing the pre-registered specificity check: no cross-scheme or random plant speeds formation, so the effect is target-specific rather than an artifact of initialization scale. One result surprised us: assist-reg toward the default slows formation, monotonically in λ (up to 3.9× at APE λ=10, with capability damage), mirroring the QK-fingerprint paper's constraint-cost law from the opposite sign. Any global spectral field, toward or away from the default, has a search cost; what accelerates is a

arm	formation step (mean ± s.d.) RoPE	APE	final capability
free	940 ± 55	600 ± 0	at floor
algebra-init	840 ± 89 (ns)	600 ± 0 (ns)	at floor
solution-init	$\mathbf{680 \pm 45}$ (-28%)	$\mathbf{380 \pm 45}$ (-37%)	at floor; higher prev
assist-reg λ=1	1080 ± 45 (slower)	960 ± 55 (slower)	at floor
assist-reg λ=10	1380 ± 130 (slower)	2340 ± 219 (slower)	APE damaged

Table 1: Assistance-hinge formation steps (n=5; one-sided exact MWU vs. free, BH-FDR). Solution-specific initialization accelerates both schemes at rank-biserial +1.0, p=0.004; algebra-class init and regularization fields do not.

target implementation	formation-time cost of the flip by ban [9]	by assistance (ours)	selection vs. free
RoPE cos-only (non-default)	+36–52%	−**23**% (faster)	5/5 vs. 0/5
APE symmetric (non-default)	2.9×	+**30**%	5/5 vs. 0/5

Table 2: Steering vs. banning for the same implementation flip (Fisher p=0.004 each). Assistance selects the same outcome a ban forces, far more cheaply; the assist cost tracks the target's intrinsic search difficulty.

solution-specific seed, not a field.

(2) Weak assistance selects the implementation. Pushed weakly toward the non-default form, every run forms the non-default implementation: 5/5 vs. 0/5 free, in both schemes (Fisher exact p=0.004 each; zero unclassified). Under RoPE-assist every final previous-token head is cos-only (`rope_imag_frac` $\in[0.016, 0.043]$, kernel peak still at $\Delta{=}-1$) vs. 5/5 phase-carried when free (`rope_imag_frac` $\approx$0.52); under APE-assist every head is symmetric (`dir_frac` $\approx$0.012) vs. 5/5 antisymmetric when free. Capability is unchanged, and the pre-registered selection criterion is met in full. A pre-registered dissection grid (108 runs) then separates the two ingredients: the weak live regularizer is the selector and the initialization is the accelerator. Reg-only flips the implementation 5/5 in both schemes, under RoPE at zero formation cost (the formation-step multiset is identical to free), while init-only accelerates without flipping (0/5 in both schemes); the combined arm's cost decomposes accordingly (≈ 0 under RoPE, +30% under APE, carried entirely by the selection).

(3) Steering is cheaper than banning. The cost of assisted selection tracks the target's intrinsic search difficulty, and is far below the ban cost for the same outcome. Forming RoPE's cos-only implementation costs −23% under assistance (faster than free) vs. +36–52% to force it by banning the phase channel [9]; forming APE's symmetric implementation costs +30% under assistance vs. 2.9× to force it by banning antisymmetry. Assisted selection $\ll$ constrained rerouting for the same implementation flip (Table 2). The two levers also have different cost geometries. A λ-grid spanning 33× shows that the ban's cost is a plateau: the constraint binds already at λ=0.3 and the delay is λ-insensitive thereafter, because the cost is the reroute itself, a step function of whether the ban binds (an earlier apparent non-monotonicity did not replicate at n=5 and is retired). Toward-default field pressure, by contrast, grows monotonically and, under APE, superlinearly, with capability damage.

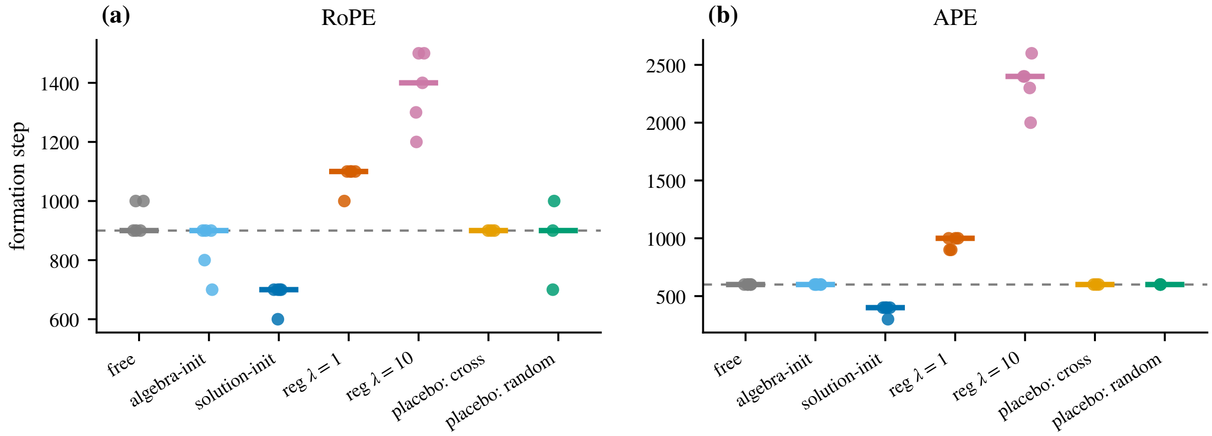

Figure 1: Formation steps by assistance arm (**a**: RoPE; **b**: APE; n=5 seeds per arm, placebos n=3; dashed line marks the free median). Solution-specific initialization accelerates both schemes; algebra-class initialization and both placebos sit at the free baseline; regularization fields slow formation, monotonically in λ.

5 Open mechanism and caveats

Scaffold, not seed. Under RoPE the planted solution is never adopted: in all five acceleration runs the planted heads end with a previous-token score near 0.02 and the circuit forms in unseeded heads. Formation nevertheless accelerates by 28%, so the RoPE plant acts as a scaffold rather than a seed; candidate mechanisms include gradient shaping through the second layer and the plant's coherent structure aiding early representation formation. Under APE the plant is adopted (5/5). A pre-registered freeze test sharpens the scaffold reading: freezing the plant entirely, with no gradient into the seeded heads, preserves and in fact doubles the speedup (500 median versus 700 unfrozen and 900 free; rank-biserial +1.0, capability intact). The seed need not even be trainable, and training's own modification of the planted heads apparently degrades the scaffold, which points to a representational or gradient-shaping mechanism rather than plant adaptation. We flag the mechanism as open rather than adjudicated. **Caveats**: n=5; 2-layer attention-only models on one synthetic task; APE formation is quantized at the 100-step eval grid (so the APE algebra-init null means "no effect at 100-step resolution," not a measured zero). These results are an existence demonstration of steerability at toy scale; the next section tests transfer.

6 Scale transfer: 160M-parameter pretraining

We repeat the two central manipulations on **Pythia-160M** (12 layers, 12 heads, 25% rotary) trained from scratch for 4B tokens through the induction formation window, n=2 seeds per cell, every contrast run in-box on the same code path.

Acceleration transfers. Installing a functional previous-token solution at initialization accelerates induction formation: the induction crossing moves from 0.364/0.388B tokens (free) to 0.267/0.267B (assisted), i.e. −27%/−31% in both seeds, at capability parity (endpoint validation gap +0.003/+0.006 nats; Fig. 2a). It also accelerates an independent language-model measure, the in-context learning crossing, by 72M/121M tokens, and it is not a measurement leak: masking the six seeded slots leaves the crossing at 0.267B, and the top induction head at crossing is an

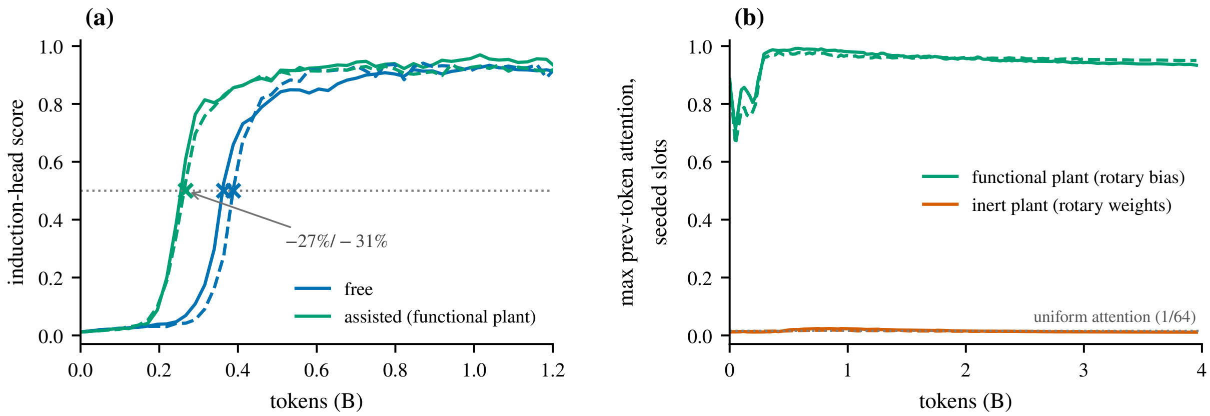

Figure 2: Acceleration transfers to 160M-parameter pretraining. **(a)** Induction-head score through the formation window: the assisted runs (green) cross the 0.5 threshold at 0.267B tokens in both seeds versus 0.364/0.388B free (blue); crosses mark the crossings. **(b)** Why the first attempt failed: the functional rotary-*bias* plant is adopted and retained (seeded-slot previous-token attention ≈0.9 throughout), whereas the earlier rotary-*weight* plant is behaviorally inert, never leaving the uniform baseline. Solid/dashed: seeds.

unseeded slot. The plant is adopted and retained throughout. This required a second attempt, and the reason is instructive: our first 160M plant was built in the rotary *weights* ($W_K{=}R(\theta)W_Q$), which realizes a content-self-match kernel that is flat at $\Delta{=}-1$; the plant is behaviorally inert, its seeded attention never leaving the uniform baseline, and it never functions as a previous-token head. Planting the same phase in the content-independent rotary *bias* gives a functional head at initialization (seeded prev-token attention 0.86), and the acceleration appears. The earlier null was a plant-construction artifact, not a scale boundary. The effect replicates on a second rotary geometry (50% rotary: −31.1%/−29.3%; architecture-level follow-up, in preparation).

Assistance selects the implementation and dominates the ban on three axes. A functional cos-only plant plus a weak population-level imag-share regularizer ($\lambda{=}3$, all heads, no ban) drives the population to the non-default implementation as deeply as the ban does: 120 and 120 of 144 heads convert to cos-only in one plant construction and 118 and 129 of 144 in the other (two seeds each), and the population-median `rope_imag_frac` is 0.006, versus the ban's 0.003 and free's 0.51. The assisted arm is also cheaper (+0.013–0.036 versus the ban's +0.039–0.048 nats) and faster (−25% to −31% versus the ban's +25% to +33%). Two independent plant constructions, four runs in total, agree.

A dose curve, and a phenomenon we name dose-defying concentration. Sweeping the regularizer over $\lambda \in \{1, 3, 10\}$ separates two populations. The population converts at the smallest dose and saturates there (median `rope_imag_frac` 0.011 → 0.006 from $\lambda{=}1$ on). The load-bearing head does the opposite: its phase content rises with pressure (0.143/0.027 at $\lambda{=}1$; 0.195/0.012 at $\lambda{=}3$; 0.326/0.244 at $\lambda{=}10$, where a second seeded head also re-retains; Fig. 3b). No λ^* exists in $[1, 10]$ and the trend excludes larger doses. A same-strength ban does convert that head (0.005), but only because a ban leaves the phase nowhere to sit. Population-mean regularization therefore cannot convert the head that carries the function: the pressure concentrates the default algebra into it. The retention half of the suppress-generic/retain-functional law is thus not passive leftover but an active, dose-defying effect, and the lever for per-head conversion is plant-side, not λ-side.

Where dominance holds. The three-axis dominance is strict at $\lambda \in \{1, 3\}$. At $\lambda{=}10$ it

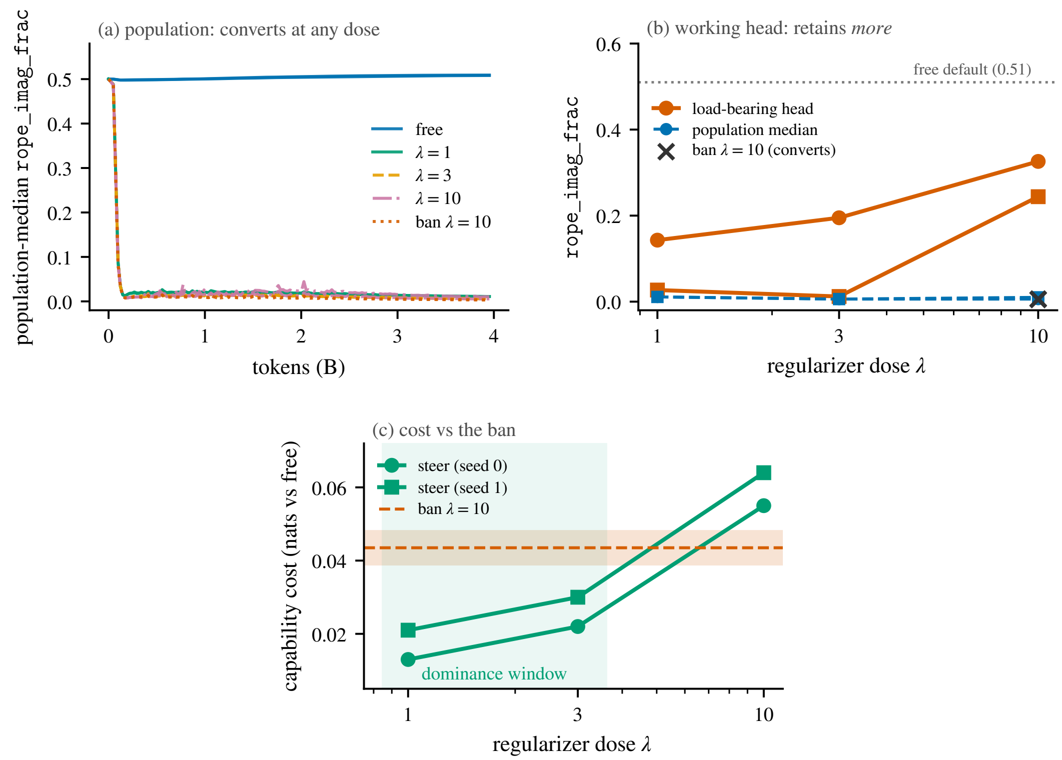

Figure 3: **Dose-defying concentration at 160M. (a)** The population converts at the smallest dose we tested and saturates there: population-median `rope_imag_frac` collapses from the free default (0.51) to the floor for every λ. **(b)** The head that carries the function does the opposite, its phase content rising with λ while the population stays at the floor. A same-strength ban does convert that head (crosses), but only because a ban leaves the phase nowhere to sit. **(c)** Cost against the ban: assistance is cheaper only within a dose window ($\lambda \leq 3$); at λ=10 the advantage inverts. Two seeds per cell (circles, squares).

breaks on cost (+0.055/+0.064 nats, above the ban's +0.039/+0.048; speed still favors assistance, −20%/−31%): assisted selection at that dose is the ban plus a plant, and the plant's working previous-token supply conflicts with maximal phase suppression.

A design caution. In the assisted models the previous-token supply is monopolized by the seeded slots: masking them collapses induction (0.013/0.014), whereas the same mask on free models leaves it intact (0.918/0.913). Assistance buys speed and selection at the price of a more fragile supply chain.

Scope. n=2 per cell, a single plant design, one model scale (160M/4B, through the formation window), and two rotary geometries; all contrasts in-box on the same code path.

7 Discussion

The constructive half of a three-way distinction. The QK-fingerprint paper separates three things the field often conflates: a trained circuit's *dependence* on a structure (post-hoc ablation is catastrophic), a *developmental preference* for it (the default forms fastest), and its *necessity*

160M arm	pop. `rope_imag_frac`	capability cost (nats)		induction speed vs. free		dominates ban?
		seed 0	seed 1	seed 0	seed 1	
free	0.51	—	—	—	—	—
assist λ=1	0.011	+0.013	+0.021	−27%	−31%	**yes**
assist λ=3	0.006	+0.022	+0.030	−27%	−25%	**yes**
assist λ=10	0.006/0.010	+0.055	+0.064	−20%	−31%	no (cost)
ban λ=10	0.003	+0.039	+0.048	+33%	+25%	—

Table 3: Steering versus banning at 160M (n=2 seeds; both values shown). Assistance matches the ban's population depth while being cheaper and faster, strictly so at $\lambda \in \{1, 3\}$. At λ=10 the cost advantage inverts, and the load-bearing head re-retains phase rather than converting: the dose-defying concentration effect.

(nothing there is strictly necessary, since bans are rerouted). That analysis is entirely negative: it prices what you forbid. The results here supply the positive half. A structure that training merely prefers can be chosen instead, and choosing is systematically cheaper than forbidding: within the dose window, assistance reaches the same implementation the ban forces while being cheaper, faster, and equally deep. Design leverage over learned algorithms lies in what you offer, not in what you prohibit.

What dose-defying concentration means. The QK-fingerprint and propagator papers report a population-level regularity: training suppresses generic spectral structure while a functional minority retains it. Read observationally, retention looks passive, a leftover after suppression. The dose curve refutes that reading. As population-mean pressure grows, the population saturates at the floor while the load-bearing head's default algebra grows, until at λ=10 a second seeded head re-retains as well. Retention is active: the pressure does not merely spare the functional head, it concentrates the default algebra into it. The law is therefore not "suppression stops at the useful head" but "suppression pushes the useful algebra into the head that needs it," which is a claim about where a resource goes, not merely about where it survives. This is what upgrades the law from observational to interventional, and it is testable elsewhere: any operator with a scarce, functionally-required channel should show the same concentration under population-level pressure.

A design principle, and a caution. Two practical statements follow. First, the levers are not interchangeable: initialization buys speed and the penalty buys selection, so per-head control must come from the plant rather than from turning λ up; the curve shows that a larger penalty backfires on exactly the head one wanted to convert. Second, assistance concentrates function into the slots you seed, which is precisely what makes it a single point of failure: masking the seeded heads collapses induction in the assisted models but not in free ones. Steering buys speed and choice at the cost of a more brittle supply chain, and any deployment of these levers should measure that.

Steering is auditable even when it hides. The implementation that steering installs is the one that evades a weight-space fingerprint (cos-only, `rope_imag_frac` ≈0.005, two orders of magnitude below the default). If weight audits were the only instrument, steering would be a way to hide an algorithm in plain sight. They are not: the propagator paper [10] finds that these same implementations remain fully visible as transient-reserve excess on the attention propagator. The weight battery reads implementation; the propagator battery reads function. A steered circuit can evade one, not both.

Limitations. At toy scale, n=5 seeds, 2-layer attention-only models, one synthetic task, and an eval grid that quantizes APE formation. At 160M, n=2 seeds per cell, a single plant design, one model scale and token budget (4B, through the formation window), and two rotary geometries;

the dose curve is three points. Every contrast is in-box on the same code path, which controls implementation drift but not the choice of task or architecture family. We have not tested a scale above 160M, a non-rotary positional scheme at scale, or a plant that targets the penalty per head rather than population-wide. The last of these is the design change our own dose curve recommends, and we have not run it.

8 Conclusion

The spectral economy of attention is not only measurable and priced [9, 10] but steerable, and the levers survive real-corpus pretraining. Weak, ban-free assistance selects which functionally-equivalent spectral implementation a circuit adopts, at a cost far below banning the alternative; solution-specific initialization accelerates circuit formation, at toy scale and again at 160M ($-27\%/-31\%$) on two rotary geometries. Within the dose window where it holds ($\lambda \in \{1, 3\}$), steering is cheaper and faster and as deep as the ban, so "which algorithm a circuit implements" is a controllable design choice, relevant to interpretability-by-construction and to installing a chosen implementation class. Two results sharpen rather than soften this. First, the dose curve shows the lever has a limit that is mechanistic, not merely practical: population-mean pressure converts the population but concentrates the default algebra into the load-bearing head, so per-head control must come from the plant, not the penalty. Second, the implementation that steering installs is weight-audit-evading (cos-only, `rope_imag_frac` $\approx$0.005). It is nevertheless not hideable, because the propagator paper [10] finds that implementations which vanish from the weight fingerprint remain fully visible as transient-reserve excess on the attention propagator (16 of 16 previous-token heads at toy scale across the free and ban-rerouted implementations, and all four at 160M). The weight battery reads implementation, the propagator battery reads function, so a steered or rerouted circuit cannot evade both.

Several application directions follow directly from these levers. The nearest is circuit priming: a functional plant costs only a few lines of initialization arithmetic yet returned roughly thirty percent of the induction formation window at 160M, so seeding known circuits could shorten the expensive early phase of pretraining, or bring forward capabilities that are gated by late-forming circuits. A second is implementation standardization. Because weak assistance chooses which of several functionally equivalent implementations forms, a lab could canonicalize the circuit implementations its analysis tools understand best, making models interpretable by construction and letting probes and audits transfer across seeds and model revisions. A third is the audit protocol itself: reading weights for implementation and propagator dynamics for function gives a two-instrument stack under which a training-time intervention, whether a steer or a reroute, remains detectable, which matters wherever provenance of a model's internal algorithms must be verified. Finally, dose-defying concentration carries a warning for regularization-based control at large, including alignment-motivated penalties on undesired internal structure: population-level pressure does not remove the targeted property from the components that use it most, but concentrates it there, so such proposals should be evaluated per component rather than on population averages. Each of these directions inherits the scope limits stated above, and the 160M results should be read as an existence proof rather than a recipe. We release the harness, seeds, and per-run tables.

Reproducibility and pre-registration. The toy harness (`src/{h1_hinge,h1_analyze,h2_m21}.py`) extends the QK-fingerprint paper's `p10` additively, and the free arms reproduce it bit-exactly; the 160M experiments use the same code path (`src/h2_m22_*.py`, `src/h2_m31.py`). Every prediction was frozen in the released plan documents before its runs; the file labels there are P-A1 to P-A3 for the hinge and selection experiments, P-M21a to P-M21d and P-A5 for the dissection grid, P-E2 for the reserve check, and P-M22-redo, P-M31, P-M31b, and P-M31λ for the 160M transfer, selection, and dose-curve experiments. The falsified predictions (algebra-class initialization, P-A1; the λ non-monotonicity flag, P-M21c; and the per-head conversion threshold within P-M31 and the λ^* clause of P-M31λ) are reported as findings. Per-run tables are under `results/h1/` and `results/h2/`. Code and data: https://github.com/HengyuLi-Ozaki-lab/qk-spectral-fingerprint.

References

[1] Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.06205.

[2] Alberto Bietti, Vivien Cabannes, Diane Bouchacourt, Hervé Jégou, and Léon Bottou. Birth of a transformer: A memory viewpoint. In *NeurIPS*, 2023. arXiv:2306.00802.

[3] Angelica Chen, Ravid Shwartz-Ziv, Kyunghyun Cho, Matthew L. Leavitt, and Naomi Saphra. Sudden drops in the loss: Syntax acquisition, phase transitions, and simplicity bias in mlms. In *ICLR*, 2024. arXiv:2309.07311.

[4] Martin Courtois, Malte Ostendorff, Leonhard Hennig, and Georg Rehm. Symmetric dot-product attention for efficient training of BERT language models. In *Findings of ACL*, 2024. arXiv:2406.06366.

[5] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework/index.html.

[6] Huang et al. Dissecting multimodal in-context learning: Modality asymmetries and circuit dynamics in modern transformers, 2026. arXiv:2601.20796.

[7] Shafayeth Jamil and Rehan Kapadia. The routing and filtering structure of attention, 2026. arXiv:2605.18826; concurrent work.

[8] Nikita Kitaev, Lukasz Kaiser, and Anselm Levskaya. Reformer: The efficient transformer. In *ICLR*, 2020. arXiv:2001.04451.

[9] Hengyu Li. Fingerprint, not blueprint: How positional schemes set the default spectral algebra of attention, 2026. Companion paper; arXiv:2607.06621.

[10] Hengyu Li. Transient reserves, sink dampers, and the failure of eigenvalue reasoning in the attention propagator, 2026. Companion paper; arXiv:2607.09279.

[11] Beren Millidge and Sid Black. The singular value decompositions of transformer weight matrices are highly interpretable. AI Alignment Forum / Conjecture, 2022. https://www.alignmentforum.org/posts/mkbGjzxD8d8XqKHzA.

[12] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. arXiv:2209.11895.

[13] Gautam Reddy. The mechanistic basis of data dependence and abrupt learning in an in-context classification task. In *ICLR*, 2024. arXiv:2312.03002.

[14] Matteo Saponati, Pascal Sager, Pau Vilimelis Aceituno, Thilo Stadelmann, and Benjamin Grewe. The underlying structures of self-attention: symmetry, directionality, and emergent dynamics in transformer training. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.10927.

[15] Aaditya K. Singh, Ted Moskovitz, Felix Hill, Stephanie C.Y. Chan, and Andrew M. Saxe. What needs to go right for an induction head? a mechanistic study of in-context learning circuits and their formation. In *ICML*, 2024. arXiv:2404.07129.

[16] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 2024. arXiv:2104.09864 (2021).

[17] Curt Tigges, Michael Hanna, Qinan Yu, and Stella Biderman. Llm circuit analyses are consistent across training and scale. In *NeurIPS*, 2024. arXiv:2407.10827.

[18] Felipe Urrutia, Jorge Salas, Alexander Kozachinskiy, Cristian Buc Calderon, Hector Pasten, and Cristobal Rojas. Decoupling positional and symbolic attention behavior in transformers, 2025. arXiv:2511.11579.

[19] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[20] Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2211.00593.